

YAYATI MANN SINGH

yayatimannsingh7@gmail.com | +91-8800806746 | [Github](#) | [LINKEDIN](#)

PORTFOLIO

SOFTWARE ENGINEER | FULL STACK DEVELOPER

Education

- **Manipal Institute of Technology Bengaluru(MIT Blr), India : CSE Core** 2023 - 2027
- **Sri Venkateshwar International School, India: CBSE (Class XII), Aggregate: 90%** 2021 - 2022
- **Sri Venkateshwar International School, India: CBSE (Class X), Aggregate: 90%** 2019 - 2020

Skills

- **Programming Languages** : C++ | C | Java | JavaScript
- **Cloud & Backend Infrastructure** : Microservice Architecture | Distributed Systems
- **Real-Time & Event-Driven Systems** : WebRTC | Socket.IO | Apache Kafka
- **Event-Driven Backend Systems** : Spring Boot | Spring Cloud (Eureka / API Gateway / Config Server)
- **Databases** : PostgreSQL | MongoDB
- **Deployment & Automation** : Docker(basic) | Docker Compose | GitHub Actions(basic CI/CD) | AWS EC2 (basic deployment) | GCP
- **Other Skills** : RESTful APIs | Authentication & Authorization(Keycloak) | Git | OOPS | DSA | OS | CN | REST-based AI Services

Projects

MicroGuard AI ([GitHub Repository](#) | Live Website : [MicroGuard](#)) Mar'26

AI-powered microservices platform for analyzing distributed system architectures with automated risk detection and recommendations.

- Designed a **cloud-native microservices architecture** using **Java, Spring Boot, Spring Cloud** (Eureka, Config Server, API Gateway) to simulate real-world distributed backend systems with secure communication via **OAuth2 (Keycloak)**.
- Built an **event-driven AI pipeline** using **Apache Kafka** for async processing and **Mistral API** for architecture analysis, reducing direct API calls via MongoDB-cached results.
- **Containerized frontend and backend services using Docker & Docker Compose**, enabling scalable multi-container networking and reproducible deployments.
- Developed an **Angular (TypeScript) frontend** with **Google SSO**, real-time analysis results, and per-user project history backed by PostgreSQL.
- Deployed the complete system on **Google Cloud (GCP)** using **Docker & Docker Compose with Nginx reverse proxy**, enabling public access, containerized scaling, and production-style service orchestration

MeetNest ([GitHub Repository](#) | Live Website : [MeetNest](#)) Dec'25

Full-stack real-time video conferencing web application for hosting secure online meetings with live video, audio, chat, and screen sharings

- Developed a **low-latency peer-to-peer communication platform** using **MERN, WebRTC, and Socket.IO**, enabling multi-participant video/audio conferencing with real-time state synchronization.
- Implemented **room-based signaling architecture** for participant join/leave handling, screen sharing, and in-meeting chat, ensuring seamless session consistency across clients.
- Containerized frontend and backend using **Docker & Docker Compose** for multi-container networking, and automated deployments to **AWS EC2** via a **GitHub Actions CI/CD** pipeline.

Certifications & Achievements

- Certification of Completion(Web Dev) - Apna College, 2025 [View Certificate](#)
- Certification of Completion(DSA) - Apna College, 2026 [View Certificate](#)
- Solved more than 200+ problems on leetcode.